

# Robotic Arm Prototype

Kinematics, mechanical integration, and coordinated joint control

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| ROBOTICS | CAD | CONTROLS |
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## Project overview

A compact robotic-arm development project focused on the full engineering chain: defining reach and payload goals, designing joints and links, selecting actuators, solving kinematics, and validating repeatable motion.

## Engineering problem

Robotic-arm performance emerges from coupled choices in geometry, torque margin, stiffness, backlash, wiring, control timing, and end-effector requirements.

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| <b>System architecture</b> <ul style="list-style-type: none"><li>- Mechanical structure: modular links, accessible joints, and replaceable end effector.</li><li>- Actuation: stepper or servo joints chosen from torque-speed and holding requirements.</li><li>- Control: joint-space commands with limits, homing, coordinated interpolation, and stop logic.</li><li>- Software: forward kinematics first, followed by bounded inverse-kinematics experiments.</li></ul> | <b>Engineering focus</b> <ul style="list-style-type: none"><li>- Static torque estimates include link mass, payload, lever arm, and safety factor.</li><li>- Workspace and singularity checks guide link dimensions and joint limits.</li><li>- Backlash and structural compliance are treated as measured system properties.</li></ul> |
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## Verification plan

- Measure repeatability at multiple workspace points and payload conditions.
- Check joint temperature, current draw, overshoot, settling, and limit behavior.
- Run pick-and-place sequences only after homing and emergency-stop tests pass.

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| <b>Status and evidence boundary</b> <p>Summer 2026 robotics design brief; final degrees of freedom and payload remain configuration-dependent.</p> |
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